ESPDI: Analytical Powered-Descent Initiation for Variable-Mass Rocket Landing

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A low-computation analytical ignition-point framework for translational fuel-efficient powered descent, with an analytical terminal-thrust-direction extension

Research Paper / Technical Report

Final experimental version

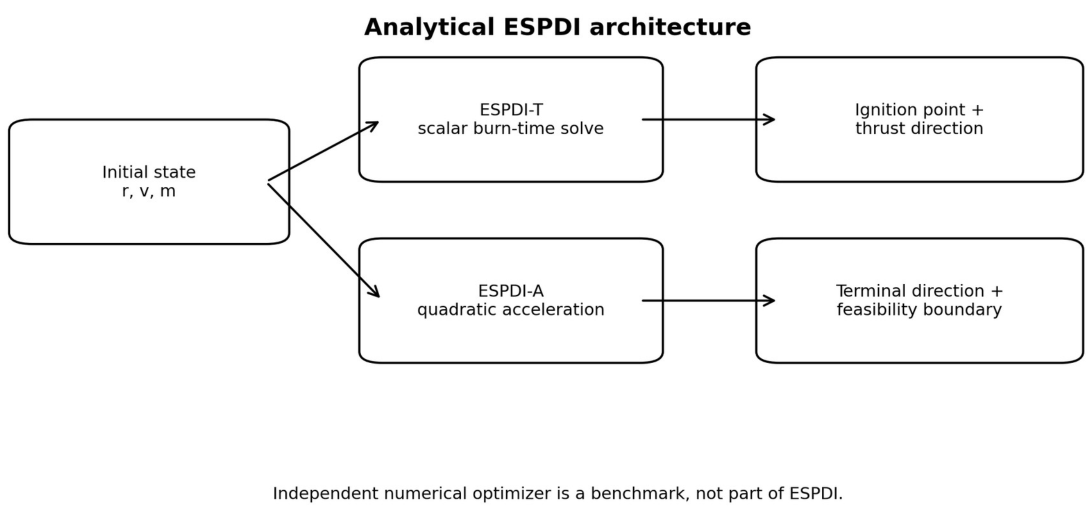

Figure 1. ESPDI research architecture used in the final study.

Abstract

Powered descent guidance for a reusable rocket can be formulated as a constrained minimum-propellant control problem. This work develops ESPDI (Enhanced Simplified Powered-Descent Initiation), an analytical method designed to calculate a recoverable maximum-thrust landing burn without solving a full online trajectory-optimization problem. The primary method, ESPDI-T, models a three-dimensional point-mass vehicle with variable mass, constant gravity, maximum thrust, and a constant inertial thrust direction. A scalar nonlinear equation determines burn duration; the terminal velocity requirement determines the thrust direction; and a closed-form position integral determines the ignition point. A finite dry-mass constraint is enforced explicitly.

ESPDI-A extends the framework by replacing the constant-vector burn with a deterministic quadratic acceleration profile constrained by terminal position, terminal velocity, and terminal thrust direction. It remains an analytical guidance construction rather than a trajectory optimizer.

The final computational study used 500 randomly generated Pair-T states for the main performance benchmark and an independent 20-state Optimizer-T validation set with two non-ESPDI starts and two meshes, plus same-seed control runs. Across all 500 translational cases, ESPDI-T and Optimizer-T used numerically indistinguishable propellant: the mean paired fuel gap was -5.28 x 10^-11 %, with 100% landing success in both methods. ESPDI-T required 0.558 s per case on average versus 13.316 s for Optimizer-T, approximately a 23.9-fold reduction in mean computation time in this implementation. Independent validation reproduced ESPDI-T from non-ESPDI starts with best-case mean gaps below 10^-6 % at both tested meshes across 20 states. For the attitude-aware extension, ESPDI-A used 3.89% more fuel than the converged Optimizer-A over 499 paired cases; one of the 500 Optimizer-A cases did not produce a benchmark result and was excluded from paired fuel-gap statistics. These results support the primary claim that ESPDI-T reproduces the numerical minimum-propellant solution for the defined idealized translational problem while requiring substantially less computation, while ESPDI-A demonstrates a useful but not globally optimal analytical extension for terminal thrust-direction control.

Research Question

Can a variable-mass analytical powered-descent initiation method determine a minimum-propellant translational landing trajectory without online trajectory optimization, and can the same analytical framework be extended to enforce a terminal thrust-direction requirement while retaining low computational complexity?

Hypotheses

- H1: Under the drag-free, constant-gravity, point-mass model with maximum thrust and no terminal attitude constraint, the optimal translational solution has a constant inertial thrust direction and is recovered by ESPDI-T.
- H2: ESPDI-T reproduces the terminal state and propellant usage of an independently implemented numerical optimizer across physically feasible three-dimensional states.
- H3: ESPDI-A can satisfy a terminal thrust-direction condition through a deterministic quadratic acceleration profile without an online trajectory optimizer.
- H4: ESPDI-A incurs a nonzero performance gap when compared with a fully optimized terminal-direction benchmark, because the analytical trajectory family is deliberately restricted.

- H5: The analytical ESPDI methods require substantially less computation than repeated nonlinear trajectory optimization for the same idealized model.

1. Problem and Motivation

A powered landing vehicle must convert a potentially large position and velocity error into a precise terminal state while respecting finite thrust and propellant. Near touchdown, a small timing error can cause a large terminal-state error. Numerical optimal-control methods can solve broad classes of this problem, but they require repeated trajectory evaluations and constrained numerical searches. ESPDI addresses a narrower problem: determine the latest maximum-thrust ignition state for a defined drag-free, point-mass landing model by exploiting its analytical structure.

The objective is not to replace all optimal-control methods. Instead, the project investigates whether a carefully derived analytical boundary can reproduce the same minimum-propellant translational solution that a numerical optimizer finds, while reducing computational cost and making the calculation predictable and interpretable.

2. Literature Context and Research Gap

Analytical powered-descent guidance has a substantial history. Apollo-style guidance represented acceleration as a low-order polynomial in time, allowing a boundary-value problem to be solved onboard. NASA literature also describes analytical dimensional-reduction approaches to fuel-optimal powered landing under constant gravity and negligible atmosphere. A representative NASA technical paper by Rea and Bishop shows that, for a bounded-thrust powered-landing subproblem, the optimal steering structure can be analytically integrated and reduced to a low-dimensional bounded search.

ESPDI is positioned in this analytical-guidance tradition but uses a specific variable-mass three-dimensional ignition-point construction. The novelty claimed here is therefore not the general existence of analytical guidance, but the explicit variable-mass 3D ignition boundary, finite dry-mass handling, direct derivation of a constant inertial thrust vector, and systematic comparison against an independently implemented optimal-control benchmark.

The research draft also adopts the interpretation that analytical powered-descent laws are generally specialized rather than universal replacements for numerical optimal control. The final results reinforce that distinction: ESPDI-T matches the unconstrained translational benchmark extremely closely, while ESPDI-A shows a measurable fuel penalty once a terminal thrust-direction constraint is imposed.

3. Novelty and Intended Contribution

- A variable-mass 3D ignition-point predictor whose primary output is an analytical recoverability boundary rather than a discretized control history.
- A direct derivation of the constant inertial thrust direction from the terminal velocity requirement.
- A closed-form position integral that converts burn duration and thrust direction into the ignition point.
- An explicit dry-mass feasibility condition included in the algorithm.
- An analytical terminal-thrust-direction extension, ESPDI-A, constructed from quadratic acceleration boundary conditions.

- An independent optimal-control benchmark with numerical feasibility checks, mesh testing, multiple initializations, and computation-time measurements.

4. System Architecture

The final system has four conceptual components. ESPDI-T is the primary analytical translational method. ESPDI-A is a secondary analytical extension. Optimizer-T and Optimizer-A are numerical benchmarks only; they are not part of ESPDI. A high-accuracy numerical integrator is used independently to verify the analytical trajectories. The visualization layer animates each method using its own state history.

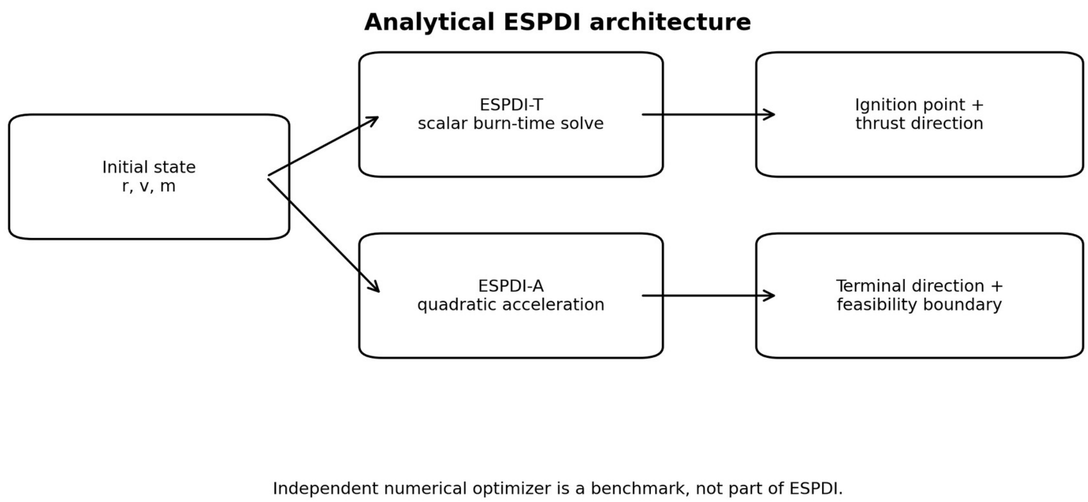

Figure 1. Final research architecture. ESPDI-T is the primary contribution; ESPDI-A is an analytical extension; the optimizer is an independent benchmark.

5. Model Assumptions

Assumption	Baseline treatment	Purpose
Gravity	Constant, downward, g = 9.81 m/s^2	Keeps the analytical derivation closed-form.
Atmospheric drag	Excluded	Isolates translational/propulsive effects.
Main-engine thrust	ESPDI-T uses T = T_max	Defines the maximum-thrust recoverability boundary.
Specific impulse	Constant, Isp = 320 s	First-order propulsion model.
Mass depletion	Continuous	Captures increasing thrust-to-mass ratio.
Dry mass	Finite, 4500 kg, enforced	Prevents nonphysical propellant depletion.
Translation	Point mass	Allows direct analytical derivation.
ESPDI-T attitude	Not constrained	Keeps the primary claim translational.
ESPDI-A attitude	Terminal thrust direction only	Adds a controlled alignment condition without a full 6DOF optimizer.

These assumptions define the scope of the claim. The project does not claim global optimality for every real rocket, atmosphere, actuator configuration, or 6DOF guidance problem.

6. ESPDI-T: Full Mathematical Derivation

6.1 State and Equations of Motion

The inertial position, velocity, mass, and thrust direction are represented by r, v, m(t), and u_hat. The baseline equations are:

$$\dot{r} = v, \qquad \dot{v} = \frac{T_{\max}}{m(t)}\hat{u} + g, \qquad g = [0, 0, -g]^T$$

Equation 1. Drag-free point-mass translational dynamics.

6.2 Variable-Mass Model

For continuous maximum-thrust operation, propellant mass flow is constant because T_max and Isp are constant:

$$\dot{m} = -\frac{T_{\max}}{I_{sp}g_0} = -\dot{m}_p, \qquad m(t) = m_0 - \dot{m}_p t$$

Equation 2. Continuous propellant depletion.

$$t_b \leq t_{prop,max} = \frac{m_0 - m_{dry}}{\dot{m}_p}$$

Equation 3. Dry-mass feasibility bound.

The dry-mass constraint is part of the solution process, not a post-processing check. A candidate burn is rejected when its terminal mass would fall below the dry mass.

6.3 Required Thrust-Generated Velocity Change

For zero terminal velocity, the integrated thrust contribution must exactly cancel the initial velocity and gravitational velocity accumulation:

$$D(t_b) = -v_0 - g t_b$$

Equation 4. Required thrust-generated velocity-change vector.

6.4 Variable-Mass Delta-v Integral

$$\Delta v_T = I_{sp} g_0 \ln\left(\frac{m_0}{m_f}\right)$$

Equation 5. Variable-mass thrust delta-v.

6.5 ESPDI-T Burn-Time Equation

Equating the available variable-mass delta-v to the magnitude of the required velocity change gives a single scalar nonlinear equation in burn time:

$$I_{sp} g_0 \ln \left[\frac{m_0}{m_0 - \dot{m}_p t_b} \right] = \| - v_0 - g t_b \|$$

Equation 6. ESPDI-T burn-time equation.

Only a scalar bounded root calculation is required. No vector-valued trajectory optimization is performed.

6.6 Constant Optimal Thrust Direction

$$\hat{u}_{ESPDI} = \frac{-v_0 - g t_b}{\| - v_0 - g t_b \|}$$

Equation 7. Constant inertial thrust direction.

For fixed thrust magnitude and a prescribed net velocity change, the triangle inequality gives ||integral d v_T|| <= integral ||d v_T||. Equality occurs when the differential thrust-generated velocity increments are collinear. Therefore, within the stated maximum-thrust translational problem, a constant inertial direction is the minimum-impulse structure. The numerical optimizer is used to test this conclusion independently rather than relying only on the analytical argument.

6.7 Ignition-Position Derivation

$$K(t_b) = I_{sp} g_0 \left[\left(t_b - \frac{m_0}{\dot{m}_p} \right) \ln \left(\frac{m_0}{m_f} \right) + t_b \right]$$

Equation 8. Closed-form position kernel.

$$r_I = r_L - v_0 t_b - K \hat{u}_{ESPDI} - \frac{1}{2} g t_b^2$$

Equation 9. ESPDI-T ignition-point equation.

For a pad located at the origin, r_L = 0. The equation directly converts the required burn duration and direction into the position from which the maximum-thrust burn reaches the pad with zero terminal velocity.

7. Translational Optimal-Control Benchmark

The benchmark optimizer solves the same idealized physical problem but is not part of ESPDI. It is allowed to select thrust magnitude, thrust direction, and final time. The continuous objective is minimum propellant:

$$J = \frac{1}{I_{sp} g_0} \int_0^{t_f} \|T(t)\| \, dt$$

Equation 10. Continuous optimal-control objective.

$$\dot{r} = v, \qquad \dot{v} = \frac{T}{m} + g, \qquad \dot{m} = -\frac{\|T\|}{I_{sp} g_0}$$

Equation 11. Benchmark dynamics and variable-mass law.

The terminal constraints are r(t_f)=0 and v(t_f)=0, with 0 <= ||T|| <= T_max and m(t_f) >= m_dry. The numerical implementation uses direct transcription with piecewise-constant thrust vectors. For a segment of duration Delta t with constant thrust vector T_k, the variable-mass equations can be integrated exactly:

$$v_{k+1} = v_k + I_{sp} g_0 \ln\left(\frac{m_k}{m_{k+1}}\right) \hat{u}_k + g\Delta t$$

Equation 12. Exact interval velocity update.

$$r_{k+1} = r_k + v_k \Delta t + K_k \hat{u}_k + \frac{1}{2} g \Delta t^2$$

Equation 13. Exact interval position update.

The use of direct thrust-vector components removes the unnecessary azimuth/elevation parameterization from the optimizer and improves numerical conditioning. Each optimization run is checked independently for terminal position, terminal velocity, final mass, and thrust-limit feasibility.

8. ESPDI-A: Analytical Terminal-Thrust-Direction Extension

ESPDI-A retains the analytical philosophy but introduces a terminal thrust-direction requirement. It does not model full quaternion dynamics, changing inertia, center-of-mass migration, gimbal rate limits, or RCS. Instead, it constructs a deterministic acceleration profile whose terminal thrust direction is prescribed.

8.1 Quadratic Acceleration Representation

$$a(\tau) = c_0 + c_1 \tau + c_2 \tau^2, \qquad \tau = \frac{t}{t_f}$$

Equation 14. Quadratic acceleration profile.

8.2 Boundary Conditions and Coefficients

With terminal position and velocity set to zero and terminal acceleration a_f prescribed, define Delta r = -r_0 - v_0 t_f and Delta v = -v_0. The boundary-value solution gives:

$$c_0 = \frac{12\Delta r}{t_f^2} - \frac{6\Delta v}{t_f} + a_f$$

Equation 15. c_0 coefficient.

$$c_1 = -\frac{48\Delta r}{t_f^2} + \frac{30\Delta v}{t_f} - 6a_f$$

Equation 16. c_1 coefficient.

$$c_2 = \frac{36\Delta r}{t_f^2} - \frac{24\Delta v}{t_f} + 6a_f$$

Equation 17. c_2 coefficient.

8.3 Terminal Thrust-Direction Condition

$$a_{T,f} = a_f - g = \frac{T_f}{m_f}\hat{b}_{3,f}$$

Equation 18. Terminal thrust-direction condition.

For an upright terminal condition, the desired body-thrust axis is aligned with the local vertical direction. The resulting condition is a terminal direction constraint, not a complete attitude controller.

8.4 Variable-Mass Coupling

$$\dot{m} = -\frac{m(t)\|a(t) - g\|}{I_{sp}g_0}$$

Equation 19. ESPDI-A variable-mass coupling.

The required thrust is T_req(t)=m(t)||a(t)-g||. The acceleration profile therefore determines the mass depletion, while mass depletion changes the required thrust margin. This coupling is handled deterministically by scalar integration and a one-dimensional feasibility search over final time.

8.5 Feasibility Calculation

- Choose a candidate final time t_f.
- Construct the quadratic coefficients analytically.
- Compute required thrust magnitude over the trajectory.
- Integrate the scalar mass equation.
- Check T_req <= T_max for the full trajectory and m >= m_dry.
- Use a one-dimensional boundary/root calculation to determine the minimum feasible t_f.

9. Attitude Interpretation and Limitations

ESPDI-A constrains terminal thrust direction as a proxy for final longitudinal alignment. It is not a full 6DOF attitude-control system. Full rotational modeling would require quaternion or rotation-matrix kinematics, angular velocity, time-varying inertia, center-of-mass migration, gimbal geometry and rates, actuator dynamics, and potentially aerodynamic moments. Those elements are intentionally outside the primary research scope.

10. Simulation Design

10.1 Three-Panel Interface

The final simulator uses a single synchronized window split into three panels. The left panel displays ESPDI-T, the center panel displays the independent optimizer for the selected comparison, and the right panel displays ESPDI-A. Each method has its own physical state history, 3D trajectory trail, moving rocket marker, landing-pad cross at (0,0,0), live position and velocity components, speed, thrust magnitude, guidance angle, fuel used, and landing status. ESPDI-T also displays its analytical ignition point. The animation does not copy one method’s trajectory into another panel.

10.2 Vehicle Specification

Parameter	Value
Initial mass	12,000 kg
Dry mass	4,500 kg
Maximum thrust	180 kN
Specific impulse	320 s
Initial T/W	1.5291
Gravity	9.81 m/s^2
Aerodynamic drag	Excluded
Model	3D point mass, variable mass

10.3 Software Architecture

- Physics layer: common gravitational and variable-mass dynamics.
- ESPDI-T: scalar burn-time root, analytical direction, analytical ignition point.
- ESPDI-A: quadratic acceleration profile and scalar feasibility boundary.
- Optimizer: independent direct-transcription nonlinear optimal-control benchmark.
- Verification: high-accuracy numerical propagation and component-wise terminal checks.
- Visualization: synchronized 3D animation with method-specific trajectories.
- Validation: independent optimizer test using fixed seeds, non-ESPDI starts, and mesh variation.

11. Experimental Methodology

11.1 Single-Case Verification

For every generated state, the analytical ESPDI result is numerically propagated through the original differential equations. Terminal position, terminal velocity, final mass, and maximum thrust are checked. The main simulator uses identical initial states within each comparison pair, so ESPDI-T and Optimizer-T solve the same physical instance; similarly ESPDI-A and Optimizer-A share their own Pair-A state.

11.2 Independent Optimizer Validation

A separate validation program reproduced the main simulator's Pair-T random-state generator from the same fixed seed and verified the first generated state matched the main simulation. The validation covered 20 Pair-T states, two meshes (N=8 and N=12), two genuinely non-ESPDI starts per mesh, and an explicit same-seed ESPDI control. This produced 120 optimization runs in total. All tested optimizer runs were physically feasible. The best independent solution across the 20 states differed from ESPDI-T by a mean of 7.95 x 10^-8 % at N=8 and 2.44 x 10^-7 % at N=12; the maximum case-wise gaps were 3.12 x 10^-7 % and 5.99 x 10^-7 %, respectively.

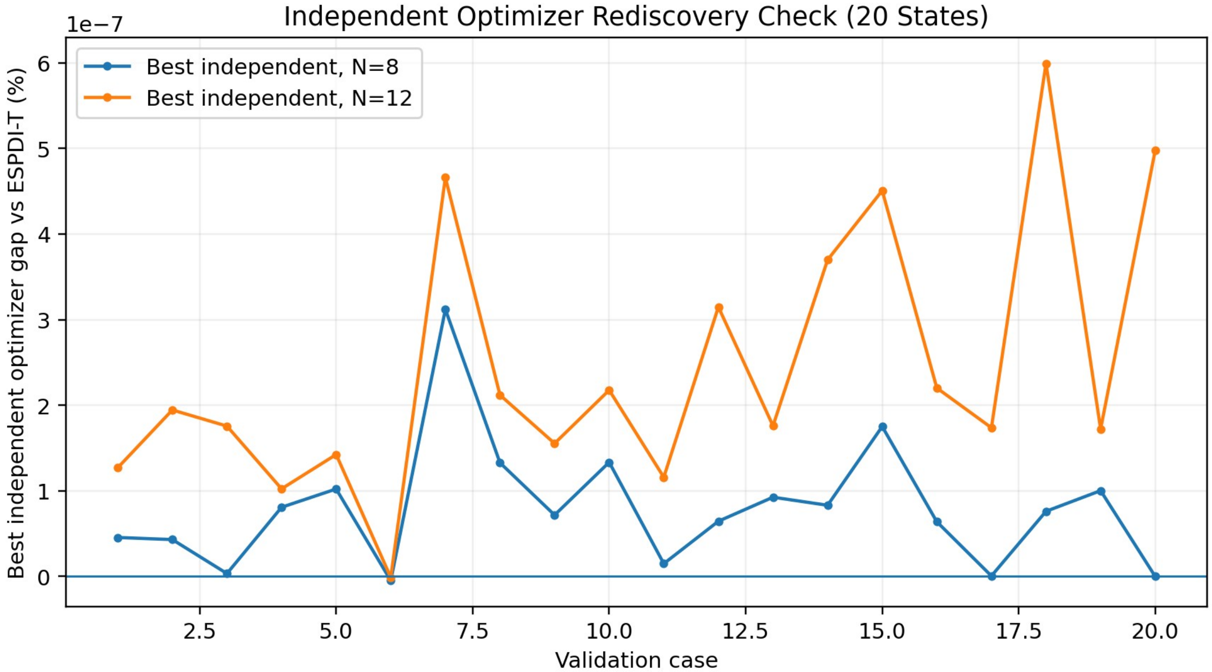

Figure 2. Independent Optimizer-T rediscovery check over the 20-state validation set. The best non-ESPDI-start solution remains within sub-micropercent fuel difference of ESPDI-T at both tested meshes.

11.3 Final Monte Carlo Comparison

The final performance dataset contained 500 Pair-T cases and 500 Pair-A cases. For each Pair-T case, ESPDI-T and Optimizer-T were run from the same initial state. For each Pair-A case, ESPDI-A and Optimizer-A were run from the same separate Pair-A initial state. The random seed was fixed for reproducibility. The Pair-T benchmark produced 500/500 successful optimizer results. Pair-A produced 499/500 optimizer results; one optimizer-A case did not yield a benchmark row and was therefore excluded from paired A fuel-gap statistics rather than being treated as a physical failure.

11.4 Statistical Metrics

- Fuel gap: (m_prop,method - m_prop,optimizer) / m_prop,optimizer x 100%.
- Burn-time error: (t_method - t_optimizer) / t_optimizer x 100%.
- Position error: ||r_f - r_target||.
- Velocity error: ||v_f - v_target||.
- Feasibility rate: fraction of cases satisfying the prescribed terminal and actuator tolerances.
- Runtime: wall-clock computation time per case on the same test environment.

12. Results

12.1 ESPDI-T vs Optimizer-T

Metric	ESPDI-T	Optimizer-T	Comparison
Cases	500	500	500 paired cases
Landing success	500/500 (100%)	500/500 (100%)	No failures
Mean propellant	874.040 kg	874.040 kg	Numerically identical
Median propellant	868.952 kg	868.952 kg	Numerically identical
Mean burn time	15.238 s	15.238 s	Numerically identical

Mean fuel gap	-	-	-5.28 x 10^-11 %
Maximum |fuel gap|	-	-	5.45 x 10^-11 %
Mean runtime	0.558 s	13.316 s	ESPDI-T ~23.9x faster
Max terminal position error	1.48 x 10^-10 m	9.00 x 10^-10 m	Numerical precision
Max terminal speed	2.83 x 10^-10 m/s	2.94 x 10^-10 m/s	Numerical precision

Across the 500 paired translational cases, the two methods produced numerically indistinguishable fuel usage. The mean paired fuel gap was -5.28 x 10^-11 %, with all 500 cases successful for both methods. The negative sign is a floating-point roundoff artifact rather than a meaningful fuel advantage. The large computational difference is the more consequential result: the analytical ESPDI-T calculation averaged 0.558 s per case, while the independent optimizer averaged 13.316 s, giving a mean runtime ratio of approximately 23.9.

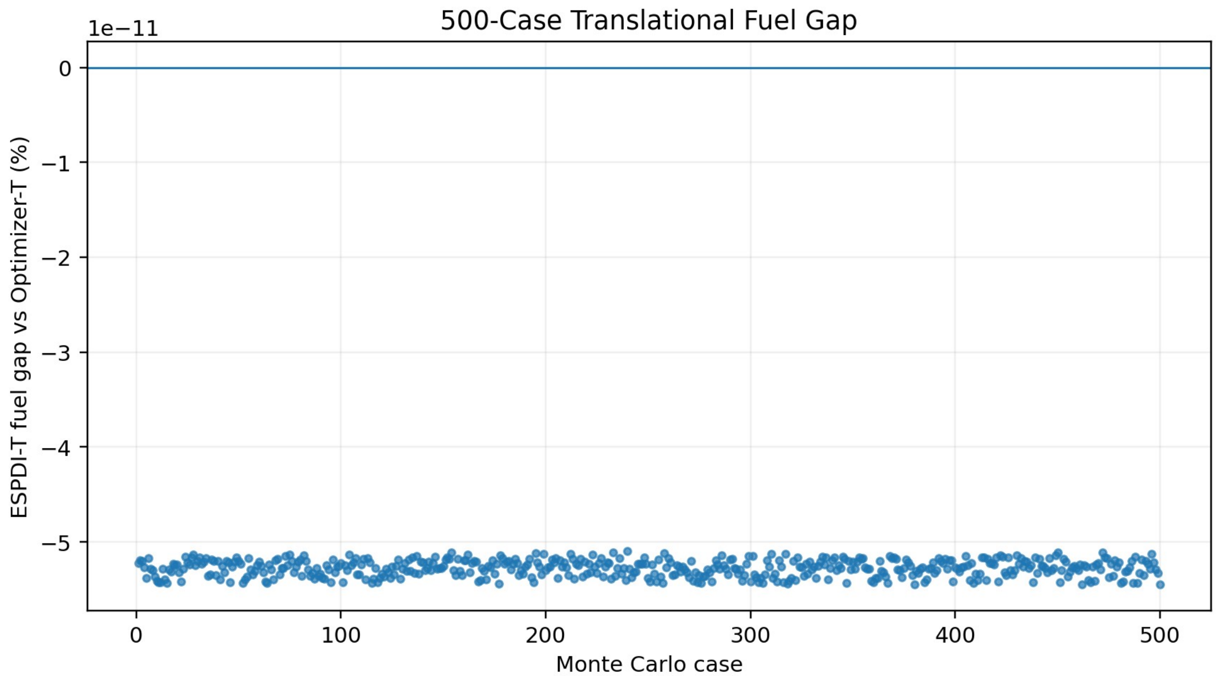

Figure 3. ESPDI-T fuel gap relative to Optimizer-T across all 500 paired cases. The plotted spread is at numerical roundoff scale.

12.2 ESPDI-A vs Optimizer-A

Metric	ESPDI-A	Optimizer-A	Comparison
Paired cases used	499	499	One optimizer-A nonconvergence excluded
ESPDI-A feasibility	500/500 (100%)	-	Analytical method feasible for all generated A states
Optimizer-A convergence	-	499/500 (99.8%)	One numerical nonconvergence
Mean propellant	1175.485 kg	1131.205 kg	ESPDI-A uses more fuel
Median propellant	1199.849 kg	1144.474 kg	-
Mean fuel gap	-	-	3.894%
Median fuel gap	-	-	3.652%
Fuel-gap range	-	-	0.221% to 7.614%
95% CI for mean gap	-	-	3.777% to 4.012%
Mean runtime	4.003 s	24.083 s	ESPDI-A ~6.0x faster

ESPDI-A shows a different behavior from ESPDI-T. The analytical terminal-direction extension uses, on average, 3.894% more propellant than the paired optimizer over 499 converged benchmark cases. The gap varies substantially by state, from 0.221% to 7.614%. At the same time, ESPDI-A is about six times faster on average in the implemented test environment. This is consistent with the intended role of ESPDI-A as an analytical extension rather than a globally optimal coupled attitude-control solution.

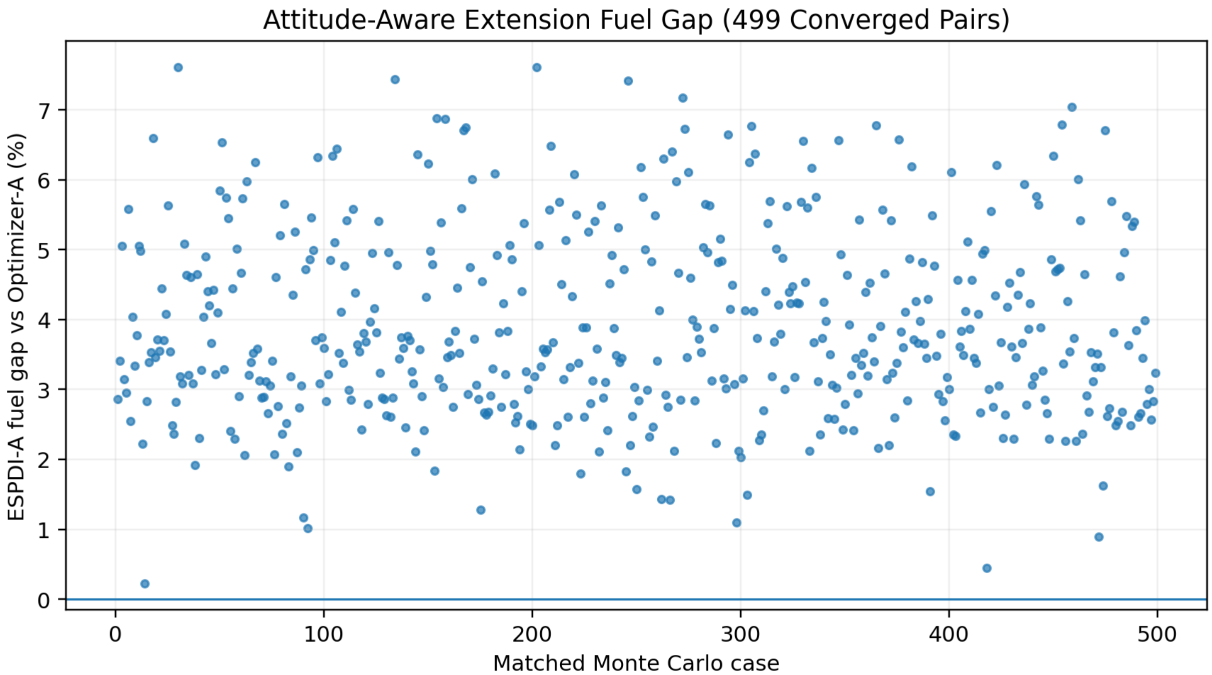

Figure 4. ESPDI-A fuel gap relative to Optimizer-A for the 499 converged paired cases.

12.3 Computation Time

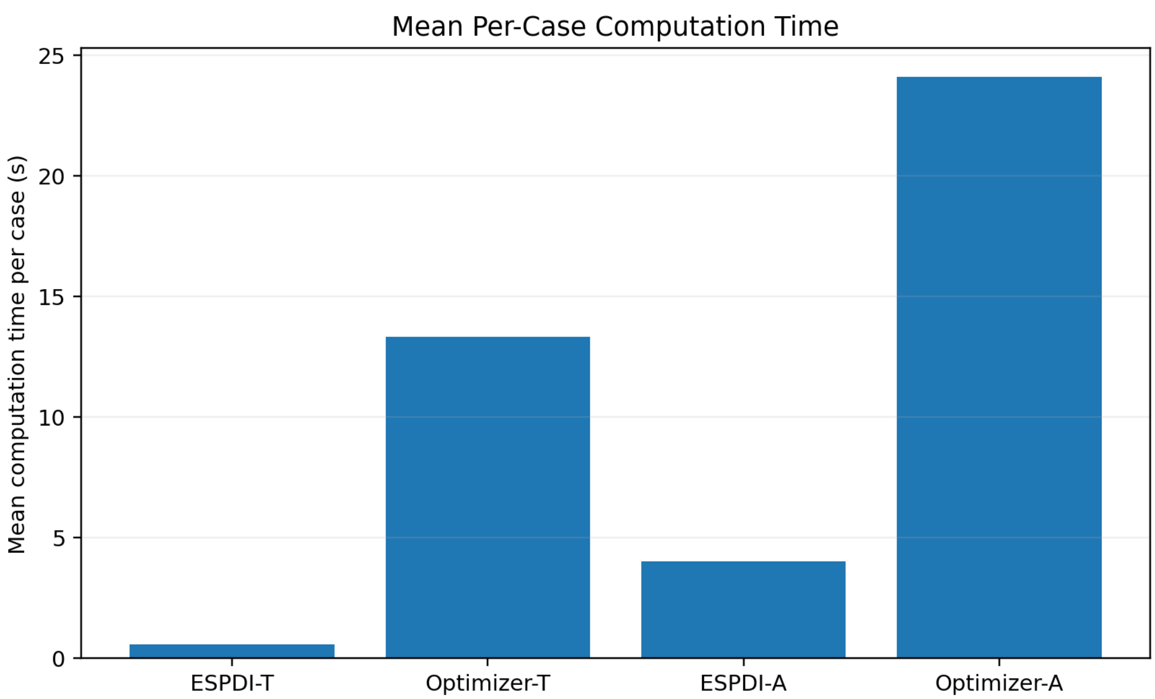

Figure 5. Mean computation time per case. Runtime is implementation- and hardware-dependent but demonstrates the computational structure of the analytical methods versus iterative optimization.

13. Discussion

The most important result is the separation between the translational and terminal-direction-constrained problems. For ESPDI-T, the 500-case Monte Carlo benchmark and the independent 20-state optimizer validation both show that the analytical solution coincides with the numerical minimum-propellant solution to far below practical modeling precision for the defined mathematical problem. The optimizer is free to vary thrust magnitude and direction, yet independent starts rediscover the same solution.

The independent validation is especially important because it addresses a potential circularity: if an optimizer were initialized with ESPDI and almost immediately stopped, identical fuel values would not establish independence. In the validation, non-ESPDI initial thrust histories converged to solutions within sub-micropercent fuel difference of ESPDI-T at both tested meshes across all 20 validation states. The same-seed control simultaneously confirmed that the benchmark can exactly represent the ESPDI trajectory.

ESPDI-A demonstrates the expected boundary of the analytical approach. Adding a terminal thrust-direction requirement removes the very simple constant-vector structure available to ESPDI-T. The deterministic quadratic profile remains cheap to compute and satisfies the terminal constraints, but it does not generally reproduce the fully optimized solution. The measured 3.894% mean fuel gap quantifies this trade-off rather than hiding it.

The computation-time results are also meaningful. ESPDI-T is approximately 23.9 times faster than the benchmark optimizer on average, while ESPDI-A is approximately 6.0 times faster than its optimizer benchmark. These ratios are not universal constants: they depend on the hardware, numerical tolerances, optimizer implementation, mesh size, and programming environment. They nevertheless demonstrate the central computational advantage of replacing a large iterative search with scalar root/feasibility calculations.

14. Limitations

- The baseline model excludes aerodynamic drag and aerodynamic moments.
- Gravity is constant and does not vary with altitude or latitude.
- The vehicle is a point mass for the translational derivation.
- Engine thrust response, ignition delay, throttling dynamics, and actuator lag are not modeled.
- ESPDI-T does not itself enforce touchdown attitude.
- ESPDI-A constrains terminal thrust direction rather than providing full 6DOF attitude control.
- The independent Optimizer-T validation covered 20 states and two meshes in the submitted validation dataset; it is a solver-trust experiment rather than a replacement for a full global-optimality proof.
- One of the 500 Optimizer-A benchmark cases did not converge and was excluded from paired A fuel-gap statistics. That case should be retained in the raw dataset and reported as a numerical benchmark limitation.
- No claim is made that ESPDI-T is globally optimal for arbitrary real-vehicle dynamics or environments.

15. Conclusion

ESPDI provides an analytical approach to a defined powered-descent landing problem in which a variable-mass point-mass rocket must reach a target with zero terminal velocity under constant gravity and maximum thrust. ESPDI-T reduces the core problem to a scalar burn-time equation, a closed-form constant thrust direction, and a closed-form ignition position. The derivation explicitly accounts for variable mass and finite dry mass.

The final 500-case translational experiment strongly supports the primary hypothesis. ESPDI-T reproduced the minimum-propellant solution of the independent numerical benchmark to numerical precision, while using approximately 23.9 times less mean computation time in the implemented environment. The independent 20-state validation further showed that non-ESPDI optimizer initializations rediscovered the ESPDI-T solution to sub-micropercent fuel differences at both tested meshes.

ESPDI-A extends the analytical framework to terminal thrust-direction control. It satisfied the prescribed terminal conditions for all generated Pair-A states but used 3.894% more fuel on average than the converged numerical benchmark. This result is not a failure of the analytical approach; it identifies the performance boundary created by adding a directional constraint without introducing a full vector-valued trajectory optimization.

The resulting research contribution is therefore specific and testable: ESPDI-T captures the special constant-vector structure of the defined unconstrained translational powered-descent problem and provides an extremely low-computation solution, while ESPDI-A demonstrates how the same analytical philosophy can be extended when additional terminal pointing structure is required.

16. Future Work

- Extend ESPDI-A to a full quaternion/variable-inertia attitude model if the terminal-direction extension proves useful.
- Introduce uncertainty in mass, thrust, Isp, and state estimation and quantify robustness.
- Add aerodynamic drag as a separate higher-fidelity experiment rather than changing the primary claim.
- Investigate terrain, glide-slope, and plume/operational constraints that might preserve analytical tractability.
- Compare ESPDI with additional modern powered-descent guidance formulations and optimized solvers.
- Publish the frozen source code, fixed seeds, raw CSV results, solver tolerances, and mesh settings for reproducibility.

17. Reproducibility and Research Integrity

The final numerical results in this manuscript were computed from the uploaded frozen-result CSV files. The main Monte Carlo dataset contains 500 Pair-T cases and 500 Pair-A analytical cases, with 500 Optimizer-T results and 499 Optimizer-A results. The independent Optimizer-T validation dataset contains 20 states, two tested meshes (N=8 and N=12), two independent non-ESPDI starts per mesh, and same-seed control runs, for 120 recorded optimization runs. Final claims should be regenerated from the frozen source code and raw CSV outputs before submission to any competition or publication venue.

18. References

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[8] NASA NTRS, “Enhancements on the Convex Programming Based Powered Descent Guidance Algorithm for Mars Landing.”

[9] Society for Science, Regeneron ISEF Grand Award Judging Criteria.

[10] Society for Science, Regeneron ISEF Categories and Subcategories.

Appendix A. Compact ESPDI-T Algorithm

- Input current r, v, m; target r_L; T_max; Isp; g; and dry mass.
- Compute m_dot_p = T_max/(Isp g0).
- Solve the scalar burn-time equation for t_b.
- Reject the state if t_b exceeds the dry-mass limit.
- Compute D = -v - g t_b and u_hat = D/||D||.
- Compute K from the closed-form position integral.
- Return t_b, u_hat, r_ESPDI, and m_f.
- Numerically propagate the resulting trajectory as an independent correctness check.

Appendix B. ESPDI-A Algorithm

- Input r, v, m, terminal position/velocity, and terminal thrust direction.
- Choose candidate t_f.
- Define terminal acceleration from gravity plus terminal thrust acceleration in the desired direction.
- Compute c_0, c_1, c_2 analytically.
- Construct a(tau) and T_req(t) = m(t)||a(t)-g||.
- Integrate the scalar variable-mass equation.
- Check T_req <= T_max and m >= m_dry.
- Find the minimum feasible t_f by scalar boundary search.
- Propagate independently for verification.

Appendix C. Final Claim Checklist

Claim	Evidence in final study
ESPDI-T is translationally correct	Independent high-accuracy propagation across 500 Monte Carlo states; terminal errors at numerical precision.
ESPDI-T matches the optimizer	500 paired benchmark cases plus independent 20-state validation with non-ESPDI starts and mesh variation.
ESPDI-A satisfies terminal direction	500 analytical Pair-A cases; all analytical trajectories satisfy terminal tolerances.
ESPDI-A is near-optimal	499 paired optimizer-A cases with a mean 3.894% fuel gap; one optimizer nonconvergence reported.
ESPDI is computationally cheaper	Mean runtime ratios of ~23.9x for T and ~6.0x for A on the tested environment.
Method scope is defined	Drag-free, constant-gravity, point-mass model with finite dry mass and stated actuator assumptions.

Simulation artifact note. The companion simulator implements the synchronized three-panel 3D animation specified in the methodology. The uploaded CSV files contain the terminal/aggregate results used here; trajectory animations are generated by the simulator itself rather than reconstructed from the aggregate Monte Carlo tables.